Stage 1 Hard Drill — With Solutions

Chapter 5.2 — Covariance, Correlation & Conditioning Laws

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — Bilinearity From a Fresh Joint Table

$X, Y \in \{0,1\}$ with joint PMF $f(0,0)$=0.3, $f(0,1)$=0.2, $f(1,0)$=0.1, $f(1,1)$=0.4. Find $\mathrm{Cov}(2X+5, 4Y-1)$.

Solution. $\mathbb{E}[X] = 0.5$, $\mathbb{E}[Y] = 0.6$, $\mathbb{E}[XY] = 1(1)(0.4) = 0.4$:

$$\mathrm{Cov}(X,Y) = 0.4 - 0.5(0.6) = 0.1$$

By bilinearity, $\mathrm{Cov}(aX+b, cY+d) = ac \cdot \mathrm{Cov}(X,Y)$:

$$\mathrm{Cov}(2X+5, 4Y-1) = (2)(4)(0.1) = \boxed{0.8}$$

⋆ *Intuition:* only the $(1,1)$ cell contributes to $\mathbb{E}[XY]$ here since $X, Y \in \{0,1\}$ — a useful shortcut whenever both variables are indicators. The affine shift $(+5, -1)$ never touches the answer; only the two scale factors $(2, 4)$ do.

Q2 — Variance of a Difference

Using Q1's table ($\mathrm{Var}(X)$=0.25, $\mathrm{Var}(Y)$=0.24, $\mathrm{Cov}(X,Y)$=0.1), find $\mathrm{Var}(X-Y)$.

Solution.

$$\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X,Y)$$

$$= 0.25 + 0.24 - 2(0.1) = \boxed{0.29}$$

▶ *Speed trick:* the *difference* formula flips the sign on the cross term versus the sum formula ($\mathrm{Var}(X+Y)$) — positively correlated legs *reduce* the variance of their difference (a spread/pairs-trade position), the opposite of what happens when you hold them together.

Q3 — Correlation From the Same Table

Using Q1/Q2's numbers, find $\rho_{X,Y}$.

Solution.

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{0.1}{\sqrt{0.25}\sqrt{0.24}} = \frac{0.1}{\sqrt{0.06}} = \boxed{\frac{1}{\sqrt{6}} \approx 0.408}$$

⋆ *Intuition:* a moderate positive ρ — these two indicators lean toward agreeing ($f(1,1) = 0.4$ is the largest single cell) but far from perfectly, consistent with ρ sitting well inside $(0,1)$ rather than at the boundary.

Q4 — A Fresh $\rho = -1$ Boundary Proof

$Y = -5X + 7$ for some random X with $\mathrm{Var}(X) > 0$. Prove $\rho_{X,Y} = -1$ regardless of X's distribution.

Solution. $\mathrm{Cov}(X,Y) = \mathrm{Cov}(X, -5X+7) = -5\,\mathrm{Var}(X)$. $\mathrm{Var}(Y) = 25\,\mathrm{Var}(X) \Rightarrow \sigma_Y = 5\sigma_X$.

$$\rho_{X,Y} = \frac{-5\,\mathrm{Var}(X)}{\sigma_X \cdot 5\sigma_X} = \frac{-5\,\mathrm{Var}(X)}{5\,\mathrm{Var}(X)} = \boxed{-1}$$

▶ *Intuition:* this holds for *any* negative slope, any distribution of X — the slope's sign alone decides ρ=+1 vs. ρ=−1; its magnitude always cancels. Never plug in numbers here — the algebra is slope-independent, so recognize the pattern $Y = aX+b$, $a<0 \Rightarrow \rho = -1$ instantly.

Q5 — Minimum-Variance Hedge, Fresh Numbers

$\mathrm{Cov}(X,Y) = 10$, $\mathrm{Var}(Y) = 5$, $\mathrm{Var}(X) = 50$. Find the hedge ratio $h^\star$ minimizing $\mathrm{Var}(X - hY)$, and the minimized variance.

Solution.

$$h^\star = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)} = \frac{10}{5} = \boxed{2}$$

$$\mathrm{Var}(X - h^\star Y) = \mathrm{Var}(X) - \frac{\mathrm{Cov}(X,Y)^2}{\mathrm{Var}(Y)} = 50 - \frac{100}{5} = \boxed{30}$$

⋆ *Intuition:* shorting 2 units of Y per unit of X cuts variance from 50 to 30 — a 40% risk reduction, limited by how imperfectly Y tracks X ($\rho^2 = \mathrm{Cov}^2/(\mathrm{Var}(X)\mathrm{Var}(Y)) = 100/250 = 0.4$ of the variance is hedgeable, matching $(50-30)/50 = 0.4$ exactly).

Q6 — Tower Property, Three Regimes

$Y \in \{0,1,2\}$ with $\mathbb{P}(Y{=}0){=}0.2$, $\mathbb{P}(Y{=}1){=}0.5$, $\mathbb{P}(Y{=}2){=}0.3$, and $\mathbb{E}[X \mid Y{=}0]{=}5$, $\mathbb{E}[X \mid Y{=}1]{=}15$, $\mathbb{E}[X \mid Y{=}2]{=}25$. Find $\mathbb{E}[X]$.

Solution.

$$\mathbb{E}[X] = 0.2(5) + 0.5(15) + 0.3(25) = 1 + 7.5 + 7.5 = \boxed{16}$$

▶ *Speed trick:* the Tower Property scales to any number of regimes without changing form — just a longer weighted sum. Note $\mathbb{E}[X]$=16 sits closest to the *middle* regime's conditional mean (15) because that regime carries the most weight (0.5).

Q7 — Wald's Identity, Fresh Numbers

A market maker executes a random number of fills N with $\mathbb{E}[N] = 8$; each fill's iid size has $\mathbb{E}[X_i] = 3.5$. Find $\mathbb{E}[S]$ where $S = \sum_{i=1}^{N} X_i$.

Solution.

$$\mathbb{E}[S] = \mathbb{E}[N] \cdot \mathbb{E}[X_i] = 8(3.5) = \boxed{28}$$

⋆ *Intuition:* identical mechanism to Example 13, new numbers — condition on N, note $\mathbb{E}[S \mid N{=}n] = n\,\mathbb{E}[X_i]$ is linear in n, then take the Tower Property outer expectation. Works regardless of what distribution N or X_i actually follow.

Q8 — Tower Property, a New Scaling

$Y \sim \mathrm{Uniform}(0,2)$, and $X \mid Y{=}y \sim \mathrm{Uniform}(0, 3y)$. Find $\mathbb{E}[X]$.

Solution. $\mathbb{E}[X \mid Y{=}y] = \frac{3y}{2}$. Then

$$\mathbb{E}[X] = \mathbb{E}\left[\frac{3Y}{2}\right] = \frac{3}{2}\,\mathbb{E}[Y] = \frac{3}{2}(1) = \boxed{\frac{3}{2}}$$

(using $\mathbb{E}[Y] = 1$ for $\mathrm{Uniform}(0,2)$).

▶ *Speed trick:* the scaling factor (here 3, versus Example 12's implicit 1) just rides straight through the Tower Property as a constant multiplier — $\mathbb{E}[X] = \frac{3}{2}\mathbb{E}[Y]$, no new integral needed once $\mathbb{E}[X \mid Y{=}y]$ is written down.

Q9 — Eve's Law on Q8's Setup

Using Q8's setup, find $\mathrm{Var}(X)$ via Eve's Law.

Solution. For $\mathrm{Uniform}(0,3y)$: $\mathrm{Var}(X \mid Y{=}y) = \frac{(3y)^2}{12} = \frac{3y^2}{4}$. For $\mathrm{Uniform}(0,2)$: $\mathbb{E}[Y^2] = \mathrm{Var}(Y) + \mathbb{E}[Y]^2 = \frac{1}{3} + 1 = \frac{4}{3}$.

$$\mathbb{E}[\mathrm{Var}(X \mid Y)] = \frac{3}{4}\,\mathbb{E}[Y^2] = \frac{3}{4} \cdot \frac{4}{3} = 1$$

$$\mathrm{Var}(\mathbb{E}[X \mid Y]) = \mathrm{Var}\left(\frac{3Y}{2}\right) = \frac{9}{4}\,\mathrm{Var}(Y) = \frac{9}{4} \cdot \frac{1}{3} = \frac{3}{4}$$

$$\mathrm{Var}(X) = 1 + \frac{3}{4} = \boxed{\frac{7}{4} = 1.75}$$

⋆ *Intuition:* direct check via $\mathbb{E}[X^2 \mid Y{=}y] = \mathrm{Var}(X \mid Y{=}y) + \mathbb{E}[X \mid Y{=}y]^2 = 3y^2 \Rightarrow \mathbb{E}[X^2] = 3\mathbb{E}[Y^2] = 4$, so $\mathrm{Var}(X) = 4 - (1.5)^2 = 1.75$ ✓— same two-route confirmation as Example 17, now with the factor of 3 threaded through both the mean and variance pieces consistently.

Q10 — Full Capstone: Tower, Eve's Law & Law of Total Covariance

Hidden regime $Y \in \{0,1\}$, $\mathbb{P}(Y{=}0){=}0.6$, $\mathbb{P}(Y{=}1){=}0.4$. Given $Y{=}0$: $(X,Z) \in \{(5,2),(9,4)\}$ equally likely. Given $Y{=}1$: $(X,Z) \in \{(12,10),(16,8)\}$ equally likely. Find (a) $\mathbb{E}[X]$, (b) $\mathrm{Var}(X)$, (c) $\mathrm{Cov}(X,Z)$.

Solution. (a) *Tower:* $\mathbb{E}[X \mid Y{=}0]{=}7$, $\mathbb{E}[X \mid Y{=}1]{=}14$:

$$\mathbb{E}[X] = 0.6(7) + 0.4(14) = \boxed{9.8}$$

(b) *Eve's Law:* $\mathrm{Var}(X \mid Y{=}0) = \mathrm{Var}(X \mid Y{=}1) = 4$ (each branch is a 2-point symmetric pair, spread ± 2) $\Rightarrow \mathbb{E}[\mathrm{Var}(X \mid Y)] = 4$.

$$\mathrm{Var}(\mathbb{E}[X \mid Y]) = 0.6(7^2) + 0.4(14^2) - 9.8^2 = 127.4 - 96.04 = 31.36$$

$$\mathrm{Var}(X) = 4 + 31.36 = \boxed{35.36 = \tfrac{394}{25}}$$

(c) *Law of Total Covariance:* $\mathbb{E}[Z \mid Y{=}0]{=}3$, $\mathbb{E}[Z \mid Y{=}1]{=}9 \Rightarrow \mathbb{E}[Z] = 0.6(3) + 0.4(9) = 5.4$. $\mathrm{Cov}(X,Z \mid Y{=}0){=}2$, $\mathrm{Cov}(X,Z \mid Y{=}1){=}{-}2 \Rightarrow \mathbb{E}[\mathrm{Cov}(X,Z \mid Y)] = 0.6(2) + 0.4(-2) = 0.4$.

$$\mathrm{Cov}(\mathbb{E}[X \mid Y], \mathbb{E}[Z \mid Y]) = \big[0.6(7)(3) + 0.4(14)(9)\big] - 9.8(5.4)$$

$$= 63.6 - 52.92 = 10.68$$

$$\mathrm{Cov}(X,Z) = 0.4 + 10.68 = \boxed{11.08 = \tfrac{262}{25}}$$

▶ *Full-circle intuition:* fresh numbers, same three-stage machine as Example 20 — Tower for the mean, Eve's Law for the variance, Law of Total Covariance for the cross-term. In every piece, the between-regime contribution (31.36 of 35.36; 10.68 of 11.08) swamps the within-regime piece, because the two regimes here are far apart (7 vs. 14, 3 vs. 9) relative to how tight each branch is (spread of only ± 2) — exactly the signature of a market dominated by regime risk rather than idiosyncratic noise.

Drill order for review: re-derive Q4 (state in one sentence why the slope's magnitude never affects ρ) and Q5 (explain why ρ^2 is exactly the fraction of variance a hedge removes) closed-book, then Q10 (the full three-law pipeline) end-to-end. Together they test whether covariance algebra, correlation, and the conditioning laws run as one connected skill, not three separate formulas.